strategy is to multiply the image with *good edges* by a constant, *k*, and add it to the image with *good contrast*. This is equivalent to convolving the original image with the 3×3 PSF shown in (d). If *k* is set to 0, the PSF becomes a delta function, and the image is left unchanged. As *k* is made larger, the image shows better edge definition. For the image in (d), a value of $k = 2$ was used: two parts of image (c) to one part of image (a). This operation mimics the eye's ability to sharpen edges, allowing objects to be more easily separated from the background.

Convolution with any of these PSFs can result in negative pixel values appearing in the final image. Even if the program can handle negative values for pixels, the image display cannot. The most common way around this is to add an offset to each of the calculated pixels, as is done in these images. An alternative is to truncate out-of-range values.

Convolution by Separability

This is a technique for fast convolution, as long as the PSF is **separable**. A PSF is said to be *separable* if it can be broken into two one-dimensional signals: a vertical and a horizontal projection. Figure 24-5 shows an example of a separable image, the square PSF. Specifically, the value of each pixel in the image is equal to the corresponding point in the horizontal projection multiplied by the corresponding point in the vertical projection. In mathematical form:

EQUATION 24-1
Image separation. An image is referred to as *separable* if it can be decomposed into horizontal and vertical projections.

$$x[r,c] = vert[r] \times horz[c]$$

where $x[r,c]$ is the two-dimensional image, and $vert[r]$ & $horz[c]$ are the one-dimensional projections. Obviously, most images do not satisfy this requirement. For example, the pillbox is not separable. There are, however, an *infinite* number of separable images. This can be understood by generating arbitrary horizontal and vertical projections, and finding the image that corresponds to them. For example, Fig. 24-6 illustrates this with profiles that are double-sided exponentials. The image that corresponds to these profiles is then found from Eq. 24-1. When displayed, the image appears as a diamond shape that exponentially decays to zero as the distance from the origin increases.

In most image processing tasks, the ideal PSF is *circularly symmetric*, such as the pillbox. Even though digitized images are usually stored and processed in the rectangular format of rows and columns, it is desired to modify the image the same in all directions. This raises the question: is there a PSF that is circularly symmetric *and* separable? The answer is, yes,